Movie Recommendations Based on Music Listening History

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Background

I am an avid consumer of both music and movies, and I particularly enjoy the Spotify platform and the recommendation system that they have built. I have a good understanding of the connections that can be recognized and leveraged between music and movies, and that will be a great help when designing this recommendation system. I have a strong background in NLP, and a great understanding of the typical models used and how to improve upon them.

Problem Statement

Music and movies are two main forms of entertainment that resonate with people and connect with their feelings and experiences. While both are generally consumed separately, they share many underlying themes, moods, and sentiments. Despite this, there is a gap in linking an individual's music preferences to potential movie recommendations. Users often find it challenging to discover movies that align with their current mood or preferences, and the vast array of movie recommendation systems primarily rely on past viewing history or generic genre-based categorization. These traditional methods may not capture the nuanced emotional or thematic preferences of a user, which are often reflected in their recent music listening habits.

Impact

By building a recommendation system which takes into account a user's recent music preferences, and uses that information to suggest a movie that may resonate with the user's current mood, we can provide a more impactful and curated media experience for the user. In a world with such large amounts of content readily available, providing a more intentional and integrated recommendation system can greatly improve people's multimedia experience.

Market

What is the size of the market? Who are the major players and main vendors in the space? What are the existing solutions?

Company Name	**Stage (startup, enterprise)**	**Product / Solution overview**	**Who is the primary customer?**	**Key differentiation vs your proposal (based on your understanding/ research)**
Spotify	Enterprise	Music & podcast recommendation system	General population	Although their music recommendation system is robust and impressive, there is no link from songs to movies.

Netflix	Enterprise	Movie & TV show recommendation system	General population	Netflix has an equally impressive recommendation system, but there is no link from TV and movies to songs the user may enjoy.
IMDB	Enterprise	Movie & TV show recommendation system	General population	Similar to Netflix, there is a great recommendation system for TV and movies, but no connection to music.

Relevant readings, market research, white papers, academic research (share title and link)

- Listening, Watching, and Reading: The Structure and Correlates of Entertainment Preferences - https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2964424/
- Finding A Formula For Movie and Music Preferences - https://npr.org/2008/12/12/98184833/finding-a-formula-for-movie-and-music-preferences

Datasets (please first identify open, public datasets or data that you believe can be generated)

- Music - Music Streaming Sessions Dataset from Spotify - https://research.atspotify.com/datasets/
- Music - The Million Playlist Dataset from Spotify - https://research.atspotify.com/datasets/
- Movies - TMDB 5000 Movie Dataset - https://kaggle.com/datasets/tmdb/tmdb-movie-metadata?select=tmdb_5000_movies.csv

Possible data science techniques and why?

- Data preprocessing (text cleaning, feature extraction, etc.)
- Sentiment analysis - to determine the general mood of a song or movie
- Summarization - extracting keywords from lyrics or movie descriptions
- Clustering - associating similar content and improving the recommendation system